EXPERIMENT ON COLD STORAGE UNIT

AIM:

- To study Vapour Compression type refrigeration cycle for cold storage.
- To find out COP of the system.
- To chill the brine solution to lower temperature.

DESCRIPTION:

Cold storage is a cabin designed to store certain goods like food stuff, fruits, vegetables and dairy products within well-defined temperature range and relative humidity. The temperature and humidity conditions are maintained inside a cold storage depend upon the type of product being stored.

The apparatus is a laboratory scale working model of a refrigeration cycle cold storage unit, portable-trolley mounted, housed on a MS square tube frame with powder-coated metallic platform to give elegant finish.

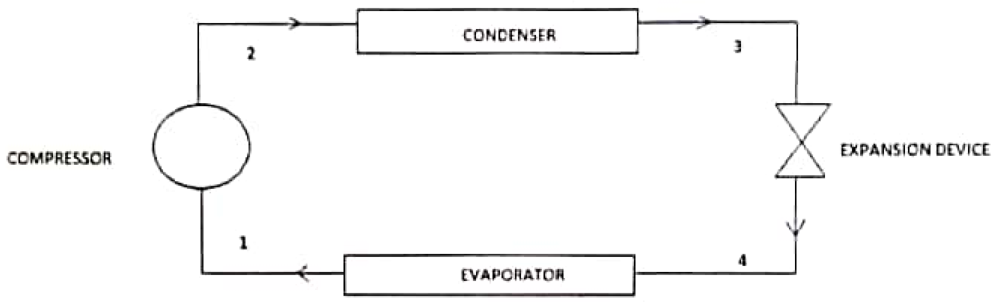
The compressor is fitted on the platform with fan-cooled condenser. The cold room/freezer - made of Galvanized iron enclosure, well insulated with bitumen and thermo cole, coiled with copper tube housed in an MS chamber duly powder coated.

THEORY:

The major difference in theory and treatment of vapour refrigeration system as compared to the air refrigeration system is that, the vapour alternatively undergoes a change of phase from vapour to liquid and liquid to vapour during the completion of a cycle. The latent heat of vaporization is utilized for carrying heat from the refrigerator, which is quite high compared with the air cycle, which depends only upon the sensible heat of air.

The substances used do not leave the plant but are circulated through the system alternatively after condensing and re-evaporating. During evaporating, it absorbs its latent heat from the brine, which is used for circulating around the cold chamber. in condensing, it gives out its latent heat to the circulating water or air of the cooler, the machine is, therefore, known as Latent Heat Pump. It absorbs its latent heat from the brine and gives out in the condenser.

All the principal parts are shown on the diagram, and path of the refrigerant flow is also shown on the diagram. The pressure is maintained at different levels in two parts of the system by the expansion valve (high side float valve). The function of the expansion valve is to allow the liquid-refrigerant under high pressure to pass at a controlled rate into the low-pressure part of the system. Some of the liquid evaporates passing through the expansion valve, but greater portion is vaporized in the evaporator at low pressure (low temperature). The liquid refrigerant absorbs its latent heat of vaporization from the air, water or other material, which is being cooled. The function of the compressor is to increase the pressure and temperature of the refrigerant above atmospheric, which will be ready to dissipate its latent heat in the condenser. In passing through the condenser, the refrigerant gives up the heat, is absorbed in the evaporator plus heat equivalent of the work done upon it by the compressor. This heat is transferred to air or water, which is used as cooling medium in the condenser.



BLOCK DIAGRAM OF VAPOR COMPRESSION CYCLE

The standard vapour compression cycle consist of following process:

1. Process- 2 represents reversible adiabatic compression from saturated vapour to the condenser pressure or superheated vapour.
2. Process 2- 3 represents reversible heat rejection at constant pressure, de- preheating and condensation.
3. Process 3- 4 represents irreversible constant enthalpic expansion from saturated liquid to the evaporator pressure.
4. Process 4- 1 represents reversible heat addition at constant pressure (evaporation to saturated vapour)

The refrigerants such as R-12, R-22, R-134a (commercially known as Freons) are used as working medium because of their properties which are required as refrigeration cycles.

Performance of standard vapour compression cycle:

Process 1- 2 is the compression process wherein mechanical work is to be supplied to a compressor. This is the quantity to be spent. Process 4- 1 represents the useful refrigeration effect. The index of performance is defined as coefficient of performance.

$$COP = \frac{\text{Useful refrigeration}}{\text{Net work}} = \frac{h1-h4}{h2-h1}$$

Thermocouple details.

T1 = Temperature of refrigerant at inlet of compressor

T2 = Temperature of refrigerant at outlet of compressor

T3 = Temperature of refrigerant at outlet of condenser

T4 = Temperature of refrigerant at outlet of expansion valve

T5 = Temperature of cold room

T6 = Temperature of specimen

PROCEDURE:

1. Put on the mains and condenser fan, respectively.
2. Now the temperature indicator will show various temperatures at various points from 1 to 6 by rotating the TSS knob.
3. Now switch on the compressor by using thermostat.
4. Allow the cold room to cool up to -10^0 C.
5. For COP calculation,
   a) Put known quantity of water/brine solution in a bowl; keep that inside the cold room.
   b) Note down the initial temperature of water/brine solution.
   c) Allow it to cool for some time, simultaneously note down the respective readings for every 15 minutes till it reach 10^0 C.
   d) Calculate COP using given formulae.

6. Depending upon the temperature and humidity conditions inside the cold room, things like vegetables, fruits, food grains etc., can also be stored in it.

Tabulation

T1	T2	T3	T4	T5	T6	P1	P2	Time	h1	h3	h2
°C	°C	°C	°C	°C	°C	Bar	Bar	s	kJ/kg	kJ/kg	kJ/kg
3.7	43.9	30.5	-18.2	17.9	13.4	1.5	10.5	900	610	625	630
-17	50	35.8	-14.6	8.8	-3.5	1.5	10.5	1800	580	452	632
-17	49.3	35.9	-14.8	4.6	-6.9	1.5	10.5	2700	580	453	634
-16.7	48	37.2	-14	2	-8.3	1.5	10.5	3600	585	455	640
-16.6	46.8	36.9	-14	0.5	-9	1.5	10.5	4500	588	454	636
-15.3	45.3	38	-13	-0.9	-9	1.5	10.5	5400	590	460	642

COP Theoretical	Q	COP carnot	time for 5 rev	W	COP actual
	kW		s	kW	
-0.75	0.007036	6.883084577	222	0.025338	0.27770062
2.461538462	0.009405	3.820895522	240	0.023438	0.40128
2.351851852	0.00706	3.861236802	244	0.023053	0.30622783
2.363636364	0.005539	3.961360124	234	0.024038	0.2304016
2.791666667	0.004528	4.044164038	230	0.024457	0.18515852
2.5	0.003774	4.252475248	225	0.025	0.15094444

CALCULATIONS:

- Theoretical COP

$$COP_{theoretical} = \frac{h1-h4}{h2-h1} = \frac{580-452}{632-580} = 2.4615$$

From 134a chart,

h_1 = Enthalpy in kJ/kg corresponding to pressure P_1 & T_1 (Vapour phase)

h_2 = Enthalpy in kJ/kg corresponding to pressure P_2 & T_2

h_3 = Enthalpy in kJ/kg corresponding to pressure P_3 & T_3

But, $h_3 = h_4$

- Actual COP

$$COP_{actual} = \frac{Q}{W} = \frac{\text{Refrigeration effect}}{\text{Compressor input}}$$

$$Q = \frac{m \cdot C\,(Ti-Tf)}{\text{Duration of test}} \text{ kW}$$

Where,

m	= Mass of water = 0.15 kg
C	= 4.18 kJ/kgK
Tf	= Final chilled water temperature
Ti	= Initial water temperature

$$Q = \frac{0.15 \cdot 4.18 \cdot (23.5-(-3.5))}{1800} = 0.00941 \text{ kW}$$

$$W = \frac{n \cdot 3600}{t \cdot EMC} = \frac{5 \cdot 3600}{240 \cdot 3200} = 0.0234 \text{ kW}$$

Where,

n	= no. of revolutions on energy meter
t	= time taken
EMC	= energy meter constant = 3200 rev/kWh

$$COP_{actual} = \frac{0.00941}{0.0234} = 0.40128$$

- Relative COP

$$COP_{relative} = \frac{COPactual}{COPtheoretical} = \frac{0.40128}{2.4615} = 0.163022$$

RESULTS:

a) **Theoretical COP = 2.4615**
b) **Actual COP = 0.40128**
c) **Relative COP = 0.163022**

CONCLUSION:

The COP obtained is very low. Apart from the low efficiency of the system, this may be due to not considering the cooling of air within the cold storage device. Also the convective heat transfer co-efficient within the refrigerator space may be low due to the absence of any flow inducing equipment.

EXPERIMENT ON SPLIT TYPE AIR CONDITIONER

AIM

To conduct performance test on air conditioner trainer (split type)

STATE OF ART

The air conditioning is that branch of engineering science which deals with the study of conditioning of air and maintaining desirable internal atmospheric condition for human comfort, irrespective of external conditions.

The four important factors for comfort air conditioning are,

- Temperature of air
- Humidity of air
- Purity of air
- Motion of air

The system which effectively controls these conditions to produce the desired effects upon the occupants of the space, is known as an air conditioning system.

Following are the main equipments or parts used in an air conditioning system

- Circulation fan
- Air conditioning unit
- Supply duct
- Supply outlets
- Return outlets
- Filters

The main classifications of air conditioning systems are the following

1. **Comfort air conditioning system**

   In comfort air conditioning, the air is brought to the required dry bulb temperature and relative humidity for the human health, comfort and efficiency. If sufficient data of the required condition is not given, then it is assumed to be 21°C dry bulb temperature and 50% relative humidity.

The comfort air conditioning may be adopted for homes, offices, shops, and restaurants, theatres, hospitals, schools etc,...

2. **Industrial air conditioning systems**

   It is an important system of air conditioning these days in which the inside dry bulb temperature and the relative humidity of the air is kept constant for proper working of the machines and for the proper research and manufacturing processes. Some of the sophisticated electronic and other machines need a particular dry bulb temperature and relative humidity.

3. **Winter air conditioning system**

   In winter air conditioning the air is heated, which is generally accompanied by humidification. The schematic arrangement of the system is shown in figure.

4. **Summer air conditioning system**

   It is the most important type of air conditioning, in which the air is cooled and generally dehumidified. The schematic arrangement of the typical summer air condition system is shown in figure.

5. **Year-Round air conditioning system**

   The year-round air conditioning system should have equipments for both the summer and winter air conditioning.

6. **Unitary air conditioning system**

   In this system, factory assembled air conditioners are installed in or adjacent to the space to be conditioned. The unitary air conditioning systems are of two types

   - Window unit.
   - Vertical packed units.

7. **Central air conditioning system**

This is the most important type of air conditioning system, which is adopted, when the cooling capacity required is 25 TR or more. The central air conditioning system is also adopted when the air flow is more than 300m^3/min or different zones in a building are to be conditioned.

The system usually works on vapour compression refrigeration cycle

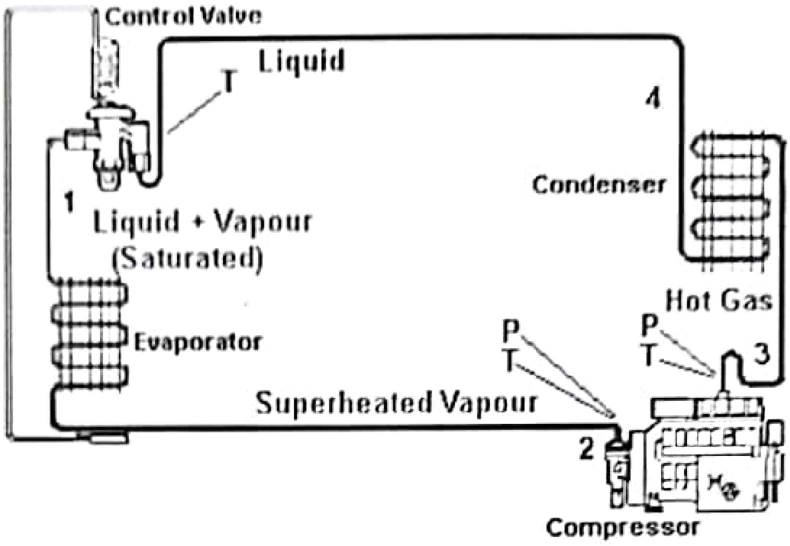


Fig. 1: Block Diagram of Vapour Compression Cycle

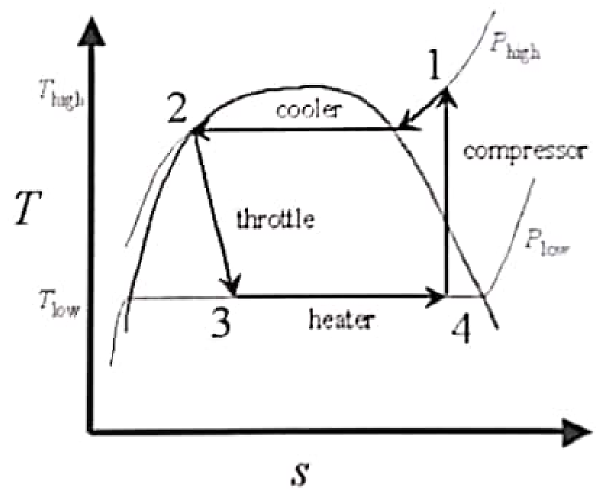


Fig. 2 T-S diagram of vapour compression cycle

Processes as numbered in the T-S Diagram.

1. Process 4-1: represents reversible adiabatic compression from saturated vapour to the condenser pressure (superheated vapour) inside the compressor (usually hermetically sealed).

2. Process 1-2: represents reversible heat rejection at constant pressure (in actual case there will be a small reduction in pressure due to pipe friction), de-superheating and condensation inside the condenser coils.
3. Process 2-3: represents irreversible constant enthalpy expansion from saturated liquid to the evaporator pressure inside the expansion valve.
4. Process 4-1: represent reversible heat addition at constant pressure (evaporation to saturated vapour) inside the evaporator.

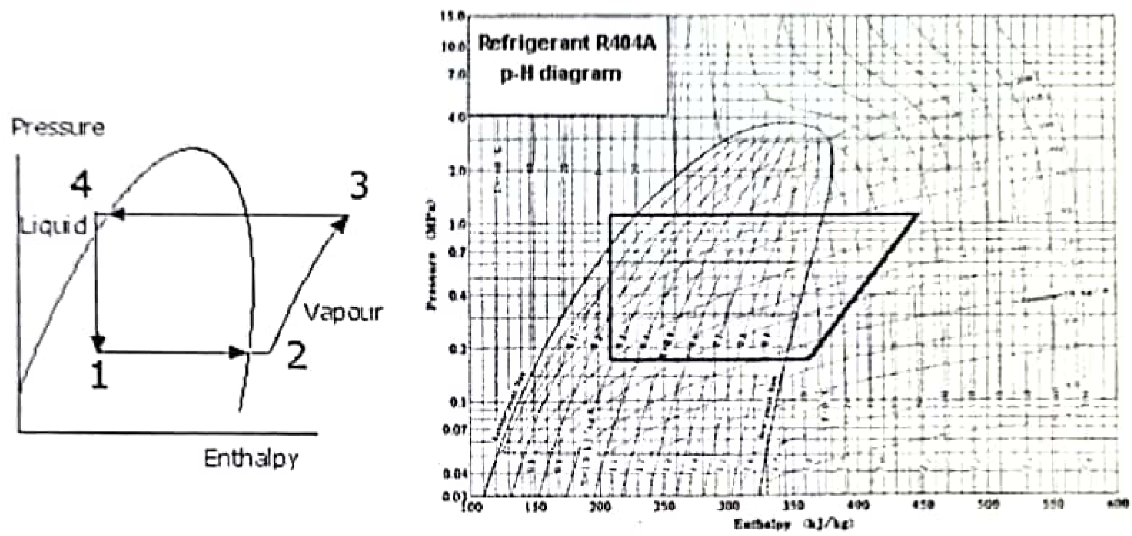


Fig. 3: P-h diagram of vapour compression cycle

SYSTEM COMPONENTS

The test rig consists of split type air conditioner unit of 1 ton capacity.

- Hermetically sealed compressor(1 ton capacity)
- Air cooled condenser
- Fan motor with blade($1/10^{th}$ / $1/8^{th}$)
- Expansion device-Capillary
- Rotameter (R-22) PG-5
- Hand shut off valve-3/8"
- Filter-Drier
- Digital Voltmeter & Digital Ammeter
- Thermostat [Electronic]
- Pressure/compound gauges
- Digital temperature indicator. K-Type(-50^0 to $+150^0$C)
- Thermocouples (K type-Cr/Al)
- DP switch for mains

- Refrigerant R-22
- Density of R-22=1172.48 at liquid state

Thermocouple details

T_1=temperature of refrigerant @ inlet of compressor.

T_2=temperature of refrigerant @ outlet of compressor.

T_3=temperature of refrigerant @ outlet of condenser.

T_4=temperature of refrigerant @ outlet of expansion device.

T_5=temperature of refrigerant @ outlet of evaporator.

PROCEDURE

1. Plug in the main chord of the system.
2. Switch ON the DP switch.
3. Press the power ON-OFF button provided on the remote control, and set the required temperature and select the mode 'cool'.
4. Now the AC will automatically switched ON depending on the ambient and the set temperature.
5. Allow air flow through the air conditioning chamber and let it stabilize for 10 to 20 minutes.
6. Record T_1, T_2, T_3, T_4, T_5, P_1, P_2, rota meter readings. Also note down wet and dry bulb temperature of ambient and outlet.
7. Take 3 to 4 readings for every 10 minutes interval.
8. Switch OFF the unit and then switch OFF the main.

EQUATIONS USED

- Theoretical COP:

$$\mathrm{COP} = \frac{h_1-h_2}{h_2-h_1}$$

Where,

h_1= Enthalpy in kJ/kg corresponding to P_1 and T_1

h_4= Enthalpy in kJ/kg corresponding to P_2 and T_3

h_2= Enthalpy in kJ/kg corresponding to P_2 and T_2

➢ Carnot COP:

$$COP = \frac{T_1}{T_2 - T_1}$$

Where,

T_1=Temperature of refrigerant @ inlet of compressor in ^oK

T_2=Temperature of refrigerant @ outlet of compressor in ^oK

➢ Mass of refrigerant:

$$m_{(ref)} = \rho_{(ref)} \times V_{(ref)}$$

Where,

$\rho_{(ref)}$= Density of the refrigerant at condenser outlet temperature in kg/m^3

$V_{(ref)} = \frac{\text{Rotameter reading} \times 10^{-3}}{60}\ m^3/s$

➢ Heat absorbed by the refrigerant (Refrigeration Effect):

$$Q = m_{(ref)} \times c_p \times (T_5 - T_4)$$

Where,

Specific heat of liquid refrigerant, c_p = 1280J/kgK

T_4=temperature of refrigerant @ outlet of expansion device

T_5=temperature of refrigerant @ outlet of evaporator

➢ Work done by the compressor:

$$W = \frac{3200 \times n}{t}$$

Where,

t = time taken for n blinks in seconds

➢ Actual COP

$$COP = \frac{\text{refrigeration effect}}{\text{work done by the compressor}}$$

➢ Relative COP

$$COP = \frac{\text{actual COP}}{\text{therotical COP}}$$

OBSERVATIONS AND TABULATIONS

P_1 (kg/cm^2)	P_2 (kg/cm^2)	T_1 (°C)	T_2 (°C)	T_3 (°C)	T_4 (°C)	T_5 (°C)	Rotameter reading (Lpm)
2.8	13.5	22.1	61.2	35.5	9.3	22.6	0.69
3.2	13.9	20.7	76.4	36.4	11	21.4	0.62
3.4	14.1	20.8	79.7	36.6	11.3	28.9	0.6
3.5	14.1	21.1	84.4	36.6	11.7	20	0.6

h_1 (kJ/kg)	h_2 (kJ/kg)	h_3 (kJ/kg)	Theoretical COP	Carnot COP	Mass flow rate of the refrigerant (kg/s)
240	275	86	4.4	7.54	0.013
242	280	88	4.05	5.27	0.012
243	282	90	3.92	4.98	0.011
244	287	90	3.58	4.64	0.011

Time taken for 5 blinks (S)	Heat absorbed by the refrigerant (W)	Work done by the compressor (W)	Actual COP	Relative COP
5.28	221.312	3030.30	0.07	0.015
5.51	159.744	2903.81	0.05	0.012
5.11	247.808	3131.11	0.04	0.010
5.41	116.864	2957.48	0.08	0.022

SAMPLE CALCULATION

- Theoretical COP:

$$COP = \frac{h_1 - h_4}{h_2 - h_1}$$

Where,

$h_1 = 243$ kJ/kg

$h_4 = 90$ kJ/kg

$h_2 = 282$ kJ/kg

$$COP = \frac{243-90}{282-243} = 3.92$$

- Carnot COP:

$$COP = \frac{T_1}{T_2 - T_1}$$

Where,

$T_1 = 20.8 + 273 = 293.8K$

$T_2 = 79.7 + 273 = 352.7K$

$$COP = \frac{293.8}{352.7-293.8} = 4.98$$

- Mass of refrigerant:

$$m_{(ref)} = \rho_{(ref)} \times V_{(ref)}$$

Where,

$\rho_{(ref)} = 1172.48$ kg/m^3

$V_{(ref)} = \frac{0.6\times10^{-3}}{60} = 10^{-5}$ m^3/s

$$m_{(ref)} = 1172.48 \times 10^{-5} = 0.011 kg/s$$

➢ Heat absorbed by the refrigerant (Refrigeration Effect):

$$Q = m_{(ref)} \times c_p \times (T_5 - T_4)$$

Where,

Specific heat of liquid refrigerant, c_p = 1280J/kgK

T_4 = 11.3 + 273 = 284.3K

T_5 = 28.9 + 273 = 301.9K

$$Q = 0.011 \times 1280 \times (301.9 - 284.3)$$

$$= 247.808 \text{ W}$$

➢ Work done by the compressor:

$$W = \frac{3200 \times n}{t}$$

Where,
n = 5
t = 5.11s

$$W = \frac{3200 \times 5}{5.11}$$

$$= 3131.11 \text{ W}$$

➢ Actual COP

$$COP = \frac{\text{refrigeration effect}}{\text{work done by the compressor}}$$

$$COP = \frac{247.808}{3131.11}$$

$$= 0.04$$

➢ Relative COP

$$COP = \frac{\text{actual COP}}{\text{therotical COP}}$$

$$COP = \frac{0.04}{3.92}$$

$$= 0.010$$

RESULT

- Theoretical COP = 3.98
- Carnot COP = 5.6
- Actual COP = 0.04
- Relative COP = 0.010

INFERENCE

From the observations and tabulations, it is clear that Theoretical COP decreases with increase in compressor outlet temperature. Theoretical COP is always less than Carnot COP because of the various kinds of losses occurring in the system.

REFERENCES

- A text book of Refrigeration and Air conditioning by R. S. Khurmi

Air Conditioner – window type

1.0 Aim: To conduct performance test on window type air conditioner.

2.0 Components and its description:

2.1 Hermetically sealed compressor: Compressors used in refrigeration systems are often described as being either hermetic, open or semi-hermetic, to describe how the compressor andmotor drive are situated in relation to the gas or vapour being compressed. In hermetic the compressor and motor driving the compressor are integrated, and operate within the pressurized gas envelope of the system. The motor is designed to operate in, and be cooled by, the refrigerant gas being compressed.It uses a one-piece welded steel casing that cannot be opened for repair; if the hermetic fails it is simply replaced with an entire new unit. The primary advantage of a hermetic is that there is no route for the gas to leak out of the system. Open compressors rely on either natural leather or synthetic rubber seals to retain the internal pressure, and these seals require a lubricant such as oil to retain their sealing properties. A hermetic system can sit unused for years, and can usually be started up again at any time without requiring maintenance or experiencing any loss of system pressure.The disadvantage is that the motor drive cannot be repaired or maintained, and the entire compressor must be removed if a motor fails. A further disadvantage is that burnt out windings can contaminate whole systems requiring the system to be entirely pumped down and the gas replaced.

2.2 Air cooled condenser: A refrigerating system component, including condenserfans, that condenses refrigerant vapour by rejecting heat to air mechanically circulated over its heat transfer surface, causing a temperature rise in the air. Desuperheating and subcooling of the refrigerant may occur as well.

2.3 Expansion device- Capillary tube: Capillary tube is one of the most commonly used throttling devices in the refrigeration and the air conditioning systems. The capillary tube is a copper tube of very small internal diameter. It is of very long length and it is coiled to several turns so that it would occupy less space. The internal diameter of the capillary tube used for the refrigeration and air conditioning applications varies from 0.5 to 2.28 mm (0.020 to 0.09 inches). Capillary tube used as the throttling device in the domestic refrigerators, deep freezers, water coolers and air conditioners.

2.4 Rotameter: A rotameter is a device that measures the flow rate of liquid or gas in a closed tube. It belongs to a class of meters called variable area meters, which measure flow rate by allowing the cross-sectional area the fluid travels through to vary, causing some measurable effect. A rotameter consists of a tapered tube, typically made of glass with a 'float', actually a shaped weight, inside that is pushed up by the drag force of the flow and pulled down by gravity. Drag force for a given fluid and float cross section is a function of square of flow speed only. A higher volumetric flow rate through a given area increases flow speed and drag force, so the float will be pushed upwards. However, as the inside of the rotameter is cone shaped (widens), the area around the float through which the medium flows increases, the flow speed and drag force decrease until there is mechanical equilibrium with the float's weight, which is calibrated with the different flow rates.

2.5 Filter- Drier: Filter is used to remove the dust particles and other unwanted foreign particles present in the outside air, while the dryer is used to remove the moisture content present in the air.

2.6 Digital voltmeter: Digital voltmeters (DVMs) are usually designed around a special type of analog-to-digital converter called anintegrating converter. Voltmeter accuracy is affected by many factors, including temperature and supply voltage variations. To ensure that a digital voltmeter's reading is within the manufacturer's specified tolerances, they should be periodically calibrated against a voltage standard.

2.7 Digital ammeter: This is a type of analog to digital converter used for displaying the amount of current passed through the circuit.

2.8 Thermostat: A thermostat is a component of a control system which senses the temperature of a system so that the system's temperature is maintained near a desired *set-point*. The thermostat does this by switching heating or cooling devices on or off, or regulating the flow of a heat transfer fluid as needed, to maintain the correct temperature. Itis used as a control unit for a heating or cooling system or a component part of a heater or air conditioner.

2.9 Pressure/ compound gauges: It is used for measuring the pressure inside a flow. Usually bourdon tube pressure gauge is used for the purpose. It is calibrated in such a way that the pressure of the fluid pushes the diaphragm inside the bourdon tube, which deflects the tube, hence making the pinion to rotate therby the needle is deflected to show the pressure reading of the fluid on the calibrated scale.

2.10 Thermocouples: Thermocouple is a temperature-measuring device consisting of two dissimilar conductors that contact each other at one or more spots. It produces a voltage when the temperature of one of the spots differs from the reference temperature at other parts of the circuit. Thermocouples are a widely used type of temperature sensor for measurement and control, and can also convert a temperature gradient into electricity. Commercial thermocouples are inexpensive, interchangeable, are supplied with standard connectors, and can measure a wide range of temperatures. In contrast to most other methods of temperature measurement, thermocouples are self-powered and require no external form of excitation. The main limitation with thermocouples is accuracy; system errors of less than one degree Celsius (°C) can be difficult to achieve.

2.11 Refrigerant R-22: It is also known asmonochlorodiflurومethane.This colourless gas is better known as **HCFC-22**, or **R-22**. It is commonly used as a propellant and refrigerant. These applications are being phased out in developed countries due to the compound's ozone depletion potential (ODP) and high global warming potential (GWP), although global use of R-22 continues to increase because of high demand in developing countries.

Chemical structure of R-22

2.12 DP switch for mains: For opening and closing the power supply

2.13 Digital temperature indicator: Displays the temperatures. The voltage produced by the thermocouples is calibrated against the known temperature, so that the temperature indicator displays the temperature according to the voltage produced by the thermocouple.

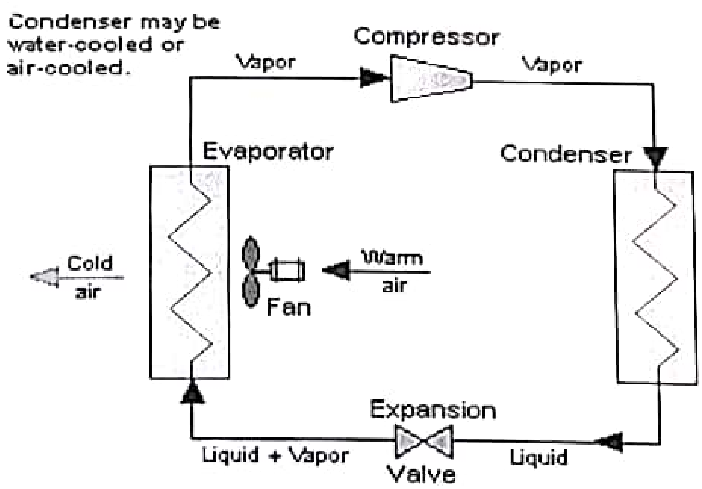
2.14 Hand shut off valves

2.15 Fan motor

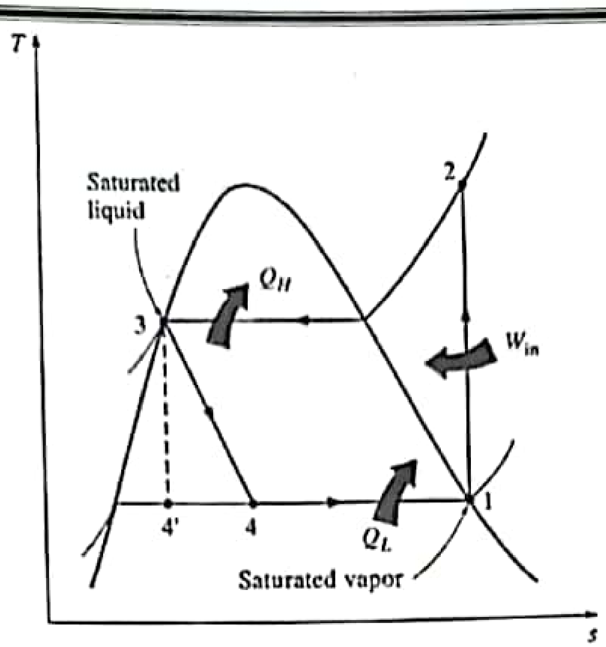
This air conditioning is a laboratory model with total instrumentation mounted on a portable trolley made of MS square tubes. The panel and the base are also of MS.

3.0 Theory:

This system works on vapour compression cycle



Block diagram of a simple vapour compression cycle



The standard vapour compression cycle consists of following processes:

Process 1-2: reversible adiabatic compression

Process 2-3: reversible heat rejection at constant pressure

Process 3-4: isenthalpic expansion

Process 4-1: reversible heat addition at constant pressure

4.0 Air conditioning

Air conditioning (often referred to as aircon, AC or A/C) is the process of altering the properties of air (primarily temperature and humidity) to more favourable conditions. More generally, air conditioning can refer to any form of technological cooling, heating, ventilation, or disinfection that modifies the condition of air.An air conditioner is asystem designed to change the air temperature and humidity within an area (used for cooling and sometimes heating depending on the air properties at a given time). The cooling is typically done using a simple refrigeration cycle, but sometimes evaporation is used, commonly for comfort cooling in buildings and motor vehicles. In construction, a complete system of heating, ventilation and air conditioning is referred to as "HVAC".

The air conditioning systems are broadly classified into two groups:

4.1 Comfort air conditioning

The comfort air conditioning systems are sub divided into three groups:

4.1.1 Summer air conditioning: the problem encountered in summer air conditioning is to reduce the sensible heat and the water vapour content of the air by cooling and dehumidifying.

4.1.2 Winter air conditioning: the problem encountered in winter air conditioning is to increase the sensible heat and water vapour content of the air by heating and humidification.

4.1.3 Year round air conditioning: this systems assures the control of temperature and humidity of air in an enclosed space throughout the year when the atmospheric conditions are changing as per the season.

4.2 Industrial air conditioning:

It provides air at required temperature and humidity to perform a specific industrial process successfully. The design conditions are not based on the feeling of the human beings, but purely on the requirement of the industrial process.

5.0 Classification based on air conditioning systems:

5.1 Window Air Conditioner

Window air conditioner is the most commonly used air conditioner for single rooms. In this air conditioner all the components, namely the compressor, condenser, expansion valve or coil, evaporator and cooling coil are enclosed in a single box. This unit is fitted in a slot made in the wall of the room, or more commonly a window sill.

5.2 Split Air Conditioner

The split air conditioner comprises of two parts: the outdoor unit and the indoor unit. The outdoor unit, fitted outside the room, houses components like the compressor, condenser and expansion valve. The indoor unit comprises the evaporator or cooling coil and the cooling fan. For this unit you don't have to make any slot in the wall of the room. Further, present day split units have aesthetic appeal and do not take up as much space as a window unit. A split air conditioner can be used to cool one or two rooms.

5.3 Packaged Air Conditioner

An HVAC designer will suggest this type of air conditioner if you want to cool more than two rooms or a larger space at your home or office. There are two possible arrangements with the package unit. In the first one, all the components, namely the compressor, condenser (which can be air cooled or water cooled), expansion valve and evaporator are housed in a single box. The cooled air is thrown by the high capacity blower, and it flows through the ducts laid through various rooms. In the second arrangement, the compressor and condenser are housed in one casing. The compressed gas passes through individual units, comprised of the expansion valve and cooling coil, located in various rooms.

5.4 Central Air Conditioning System

Central air conditioning is used for cooling big buildings, houses, offices, entire hotels, gyms, movie theatres, factories etc. If the whole building is to be air conditioned, HVAC engineers find that putting individual units in each of the rooms is very expensive making this a better option. A central air conditioning system is comprised of a huge compressor that has the capacity to produce hundreds of tons of air conditioning. Cooling big halls, malls, huge spaces, galleries etc. is usually only feasible with central conditioning units.

6.0 Thermocouple details:

T_1 = temperature of the refrigerant at the inlet of compressor

T_2 = temperature of the refrigerant at the outlet of the compressor

T_3 = temperature of the refrigerant at the outlet of the condenser

T_4 = temperature of the refrigerant at the outlet of the expansion device

T_5 = ambient temperature

7.0 Equations used:

Theoretical COP

$$COP = \frac{h_1 - h_4}{h_2 - h_1}$$

From R-22 p-h chart,

h_1 = enthalpy in kJ/kg corresponding to P_1 and T_1

h_4 = enthalpy in kJ/kg corresponding to P_2 and T_3

h_2 = enthalpy in kJ/kg corresponding to P_2 and T_2

Carnot COP

$$COP = \frac{T_1}{T_2 - T_1}$$

Where,

T_1 = temperature of refrigerant at the inlet of the compressor in K

T_2 = temperature of refrigerant at the outlet of the compressor in K

Actual COP

$$COP = \frac{Q}{W} = \frac{Refrigeration\ Effect}{Compressor\ Input}$$

$$Q = m_a * c_{pa} * (T_i - T_o)$$

Where,

m_a = mass of air

m_a = velocity of air X area X density of air

velocity of air can be found out using the anemometer

c_{pa} = specific heat of air = 1.005kJ/kg K

T_i = inlet temperature of air in K

T_o = outlet temperature of air in K

Density of air = 1.197kg/m^3

Compressor input

$$W = \frac{V * I}{1000}$$

Where,

V = digital voltmeter reading

I = digital ammeter reading

Relative COP

$$COP\ rel. = \frac{actual\ COP}{theoretical\ COP}$$

Mass of refrigerant

$$M_{ref} = \rho_{ref} * V_{ref}$$

$$V_{ref} = \frac{rotameter\ reading}{1000 * 3600}$$

ρ_{ref}= density at T_3 from R-22 table, kg/m^3

8.0 Procedure:

i. Plug in the mains of the system

ii. Switch ON the DP switch

iii. Press the power ON-OFF button provided on the remote control and set the required temperature and select the mode cool

iv. Now the AC will automatically get switched on depending upon the ambient and set the temperature

v. Allow the air to flow through the air conditioning chamber and let it stabilize for 10-20 minutes

vi. Record T_1, T_2, T_3, T_4, T_5, P_1, P_2, rota meter readings. Also note down wet and dry bulb temperature of ambient and outlet

vii. Take 5 readings for every 10 minutes interval.

viii. Switch off the unit by pressing 'power off' button and then switch off the mains

9.0 Observations and calculations

Sl no	Pressure p1(bar)	Pressure p2(bar)	T1(°C)	T2(°C)	T3(°C)	T4(°C)	T5(°C)	Tdbt (°C)	Twbt (°C)
1	3.4	19.9	31.6	97.8	46.3	18.4	19.7	20	17
2	3.5	19	31	105.2	44.8	20.4	21.1	18	16
3	3.4	18.6	33.4	106.1	45.8	21.8	22.5	19	16
4	3.5	18.5	33.5	106.2	45.3	21.5	22	17	15

Sl no	Flow rate of refrigerant (kg/s)	Time for 5 blinking(s)	Refrigeration effect (kJ/kg)	Carnot COP	V(volt)	I (ampere)	Power(kw)
1	0.75	8.2	0.193	4.601	231	5.4	1247.4
2	0.75	8.1	0.155	4.097	230	5.32	1223.6
3	0.78	8.1	0.1768	4.214	230	5.26	1209.8
4	0.75	8.2	0.1758	4.216	231	5.25	1212.8

10.0 Results: The experiment is done and various readings are taken. The performance of the window type air conditioner is studied.

11.0 Inference: It is seen that the COP of the system is increasing with the increase in compressor pressure. But in one reading the COP is found to get reduced, which may be due to some experimental errors, the reading might have taken before the system attains the steady state. Unlike it split type system, the mass flow rate remain almost constant, so obviously the refrigerating effect will increase with the increase in the condenser pressure. On comparing with the split type system it is seen that, if the mass flow rate was made same in both the cases, split type system has more COP and produces more refrigerating effect with less compressor work. But actually the performance of the split ac system must be little bit lower than window type inspite of the other advantages like less noise, and no need of making holes for its installation. The lower performance of split type ac is because the ducts connecting between the outer unit (comprises of compressor, condenser and expansion valve) and the inside unit (evaporator and cooling fan), there will be some leakage of heat into the ducts thereby increasing the temperature of the refrigerant coming through the duct, even though these are well insulated there is a small chance of leakage. We were not able to measure the mass flow rate of air, since the anemometer was not available, therefore the refrigerating effect per unit mass of air is calculated here, and so the theoretical COP cannot be calculated.